

TAMPA EXECUTIVE, FL, USA

WMO: 722021

Lat: 28.014N

Lon: 82.345W

Elev: 22

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 92816

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.3	37.5	23.2	17.5	46.0	28.5	22.4	49.4	18.6	70.7	16.6	69.0	1.4	20	0.318

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.0	95.4	77.6	94.7	77.6	92.8	77.4	81.3	89.8	80.5	89.2	79.8	88.6	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.3	151.9	84.8	78.7	149.2	84.5	77.3	142.2	83.6	44.8	89.6	43.8	89.4	43.2	88.1	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	12.8	11.1	27.8	98.1	7.7	2.1	22.2	99.6	17.7	100.8	13.4	102.0	7.8	103.5		
(5)			26.9	82.3	6.7	1.0	22.1	83.0	18.1	83.6	14.3	84.1	9.5	84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	73.0	59.8	63.5	66.9	72.0	77.5	81.9	82.7	83.3	81.6	75.3	67.0	64.4
	DBStd	9.96	8.62	7.96	7.03	5.23	3.94	2.54	2.30	2.31	2.35	5.78	7.04	7.95
	HDD50	33	20	7	1	0	0	0	0	0	0	0	2	3
	HDD65	572	203	112	66	11	0	0	0	0	0	9	59	112
	CDD50	8446	325	383	525	660	854	956	1015	1032	949	786	512	449
	CDD65	3511	43	69	125	222	389	506	550	567	499	329	119	93
	CDH74	34068	348	591	1185	2357	4185	5150	5412	5657	4714	2836	951	682
(13)	CDH80	14444	41	101	291	847	1968	2500	2538	2617	2119	1100	200	122
(14)	Wind	WSAvg	3.6	4.2	4.2	4.4	4.2	3.9	3.0	2.3	2.6	3.2	3.6	3.9
(15) Precipitation	PrecAvg	52.3	2.9	2.6	3.4	2.6	3.0	8.1	8.1	8.3	6.9	2.8	1.7	2.4
	PrecMax	65.6	6.2	9.2	8.1	5.7	12.1	17.9	13.3	13.2	16.0	6.7	5.9	12.8
	PrecMin	37.2	0.6	0.3	0.2	0.2	0.4	1.0	4.0	5.7	2.4	0.1	0.1	0.1
	PrecStd	7.3	1.5	2.0	2.3	1.3	2.5	3.6	2.3	1.9	3.3	1.5	1.2	2.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.5	86.0	88.2	91.4	96.6	97.2	97.1	97.2	95.2	91.4	87.8	86.2
		MCWB	69.8	70.2	71.5	72.2	74.1	76.7	78.5	79.2	77.9	76.2	74.1	72.8
	2%	DB	81.2	82.2	86.0	89.8	93.2	95.1	95.1	95.2	93.0	90.4	84.3	82.4
		MCWB	68.8	69.1	69.1	71.1	73.7	76.9	78.5	78.9	77.5	75.8	71.8	71.1
	5%	DB	78.8	81.1	82.4	87.8	91.1	93.0	93.1	93.2	91.2	88.4	82.0	81.2
		MCWB	67.2	68.4	68.4	70.9	73.2	77.2	78.6	79.0	77.2	74.8	69.8	70.2
	10%	DB	74.6	78.6	81.2	85.6	90.0	90.9	91.0	91.1	90.3	86.3	80.6	78.7
		MCWB	65.7	66.7	67.5	69.4	73.0	77.0	78.5	78.8	77.2	73.7	68.6	68.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.4	74.0	77.3	76.6	79.6	81.3	82.5	82.4	81.2	80.2	77.0	75.2
		MCDB	79.7	80.2	81.6	85.5	89.1	90.5	90.6	92.1	89.0	86.9	83.9	82.1
	2%	WB	71.2	72.3	74.0	75.1	77.8	80.3	81.2	81.3	80.1	78.6	74.4	73.6
		MCDB	77.3	79.0	79.7	84.0	87.5	89.6	89.5	90.0	88.5	85.7	81.3	79.8
	5%	WB	69.3	70.6	72.1	73.7	76.5	79.3	80.2	80.6	79.2	77.6	72.4	72.1
		MCDB	75.0	77.3	78.7	82.3	86.2	88.6	88.9	89.4	87.8	84.6	78.8	78.3
	10%	WB	67.1	69.1	70.0	72.3	75.3	78.4	79.3	79.8	78.3	76.1	70.6	70.3
		MCDB	73.9	75.7	76.6	81.2	84.8	87.5	88.2	88.6	86.8	83.4	76.5	76.0
(35) Mean Daily Temperature Range	5% DB	MDBR	24.1	23.9	24.7	25.3	24.0	19.6	18.4	18.0	18.5	21.5	24.0	22.7
		MCDBR	24.9	25.0	25.5	26.8	26.1	22.4	20.4	19.9	19.9	21.5	23.8	22.4
		MCWBR	13.6	12.7	11.5	10.9	9.7	8.0	7.5	7.5	7.5	8.9	12.0	11.8
	5% WB	MCDBR	21.1	20.9	19.6	22.2	21.9	18.8	18.3	18.1	17.5	17.9	19.5	19.1
		MCWBR	13.2	12.1	11.4	11.1	9.4	7.9	7.9	7.8	7.2	8.0	11.3	11.3
(40) Clear-Sky Solar Irradiance	taub		0.341	0.354	0.367	0.389	0.425	0.451	0.475	0.456	0.421	0.389	0.354	0.340
	taud		2.517	2.492	2.442	2.346	2.304	2.309	2.244	2.314	2.383	2.451	2.496	2.547
	Ebn at Noon		286	292	293	288	276	267	260	265	272	276	279	282
	Edh at Noon		29	32	36	40	42	42	44	41	37	33	29	27
(44) All-Sky Solar Radiation	RadAvg		1119	1346	1700	2033	2146	1927	1836	1771	1614	1489	1216	1033
	RadStd		79	85	137	101	124	157	114	112	101	94	62	84

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.74	N/A	N/A	N/A	N/A	N/A	-7	-81	+284	+200

Nomenclature: See separate page